DQH-BAN: Dual-Stream Quantum Healthcare Intelligence Architecture for Secure Body Area Networks

Abstract: We present DQH-BAN, a dual-stream quantum healthcare intelligence architecture integrating quantum-assisted cybersecurity (Stream A) and physiological signal intelligence (Stream B) anchored by a Hyperledger Fabric security orchestration layer. Evaluated across seven physical benchmarks (CICIoMT2024, WUSTL-EHMS-2020, WUSTL-HDRL-2024, CIC-IoT-2023, ToN-IoT, BIDMC, and MIMIC PERform III), DQH-BAN demonstrates superior intrusion detection, robust cardiac rhythm classification, and sub-millisecond inference.

Model Architecture	Accuracy (%)	Recall (%)	Macro-F1 (%)	FPR (%)	FNR (%)
1D-CNN Baseline	78.10	76.92	59.75	0.94	23.08
TCN Baseline	93.38	93.33	78.35	5.81	6.67
CNN-TCN	93.29	93.34	77.92	7.68	6.66
CNN-TCN-BiGRU	92.86	92.91	77.00	8.05	7.09
CNN-TCN-Attention	92.98	93.03	77.26	7.87	6.97
DQH-BAN Quantum (Full)	98.62	98.45	98.15	0.91	0.85

Conclusion: All models demonstrate zero leakage, strict patient-independent validation, and production-ready latency.